ClipSAT : Math for AI & Data Science

Unit 7.01: Logistic Regression & Classification

Name: _____

Q1. In evaluating a binary classifier, the confusion matrix shows TP = 45 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 81.0%
[ ] B: 81.8%
[ ] C: 52.4%
[ ] D: 13

Q2. In evaluating a binary classifier, the confusion matrix shows TP = 50 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 81.8%
[ ] B: 83.3%
[ ] C: 54.5%
[ ] D: 13

Q3. In evaluating a binary classifier, the confusion matrix shows TP = 55 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 82.6%
[ ] B: 84.6%
[ ] C: 56.5%
[ ] D: 13

Q4. In evaluating a binary classifier, the confusion matrix shows TP = 60 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 83.3%
[ ] B: 85.7%
[ ] C: 58.3%
[ ] D: 13

Q5. In evaluating a binary classifier, the confusion matrix shows TP = 65 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 84.0%
[ ] B: 86.7%
[ ] C: 60.0%
[ ] D: 13

Q6. In evaluating a binary classifier, the confusion matrix shows TP = 70 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 84.6%
[ ] B: 87.5%
[ ] C: 61.5%
[ ] D: 13

Q7. A spam classification model identifies TP = 30 genuine spam emails and FP = 10 legitimate emails falsely flagged as spam. What is the precision of this model?

[ ] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%

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Q8. A spam classification model identifies TP = 60 genuine spam emails and FP = 20 legitimate emails falsely flagged as spam. What is the precision of this model?

[ ] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%